

Mayo Clinic: Building the Platform for the Future of Medicine

A Case Study in Healthcare AI Platform Strategy

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The Question on the Table

In the spring of 2020, Dr. John Halamka sat in his new office in Rochester, Minnesota, facing a question that would shape the next decade of American healthcare: could a 157-year-old medical institution remake itself as a technology platform — and should it?

Three months earlier, Halamka had been at Harvard Medical School, teaching what he knew about health information technology to a generation of students who had grown up with smartphones but were entering a healthcare system still faxing patient records. Then Mayo Clinic offered him something different: the chance to build an entirely new kind of healthcare organization from inside one of the most trusted brands in medicine. He said yes.

He arrived on January 1, 2020, as the inaugural President of Mayo Clinic Platform — a newly created business unit charged with turning Mayo's most valuable, and most sensitive, asset into a competitive advantage. That asset was data: decades of de-identified patient records representing conditions, treatments, and outcomes at a scale no algorithm had ever been trained on.

Then COVID-19 arrived in March 2020. The question of what a hospital could do with its data became the most urgent question in global health.

“When I look at the 40 years that I've been in this industry ... I've often looked at 'when is there a perfect storm for innovation?’”

— Dr. John Halamka — HealthLeaders Media, 2020

He believed he had found one.

Mayo Clinic: Institution, Not Just Hospital

To understand what Halamka was building, you first need to understand what Mayo Clinic is — and what it is not.

Mayo Clinic is not a hospital chain in the commercial sense. It is a nonprofit integrated medical practice founded in 1863 in Rochester, Minnesota, organized around a mission unchanged across more than fifteen decades: to inspire hope and contribute to health and well-being by providing the best care to every patient through integrated clinical practice, education, and research. The institution grew from a surgical practice started by Dr. William Worrall Mayo into the most recognized name in American medicine, built on a philosophy of physician collaboration at a time when solo practice was the norm.

By 2023, that mission had produced a \$17.9 billion enterprise — revenues up 10.2 percent from \$16.3 billion in 2022 — with an operating income of \$1.1 billion and a 6 percent operating margin. Mayo participated in 2,500 clinical trials annually, served approximately 1.3 million patients from more than 130 countries, and employed roughly 86,000 people across campuses in Minnesota, Arizona, and Florida.

But the financial figures obscured what made Mayo genuinely distinctive. Patients did not typically come to Mayo for routine care. They came because every other physician had run out of answers: the rare cancer whose histology no one recognized, the autoimmune condition that had baffled six specialists, the surgical challenge requiring a team of twenty in synchronized concert. Mayo had spent generations becoming the institution of last resort — and first resort — for medical complexity.

That diagnostic depth generated data unlike anything in commercial healthcare. Not simply claims data or routine encounter records, but longitudinal patient histories tracked across decades and specialties: genomic sequences paired with 30-year clinical trajectories, imaging studies correlated with pathology and long-term survival, the clinical notes of 35,000 physicians capturing the texture of medical reasoning in ways structured data fields never could. Other health systems had more patients. None had this depth of complexity. And it was the complexity — the rare diseases, the edge cases, the puzzles — that made Mayo's data most valuable for training AI meant to learn from the hardest problems, not merely the most common ones.

In 2019, Mayo's leadership made a consequential decision: this data could be the foundation of something entirely new. The question was how to monetize it without betraying the patients who had generated it.

John Halamka: The Right Person for an Impossible Job

By late 2019, Dr. John Halamka was one of the most recognizable figures in health information technology — and one of the most unusual. He was simultaneously a practicing emergency medicine physician, a computer scientist, a published author, and a federal policy advisor. The combination was rare enough that it made him something healthcare had long needed but rarely produced: someone who understood both what algorithms could do and what it felt like to stand at a patient's bedside at 2 a.m. making a decision under genuine uncertainty.

Born in Des Moines in 1968, he earned dual degrees in Public Policy and Medical Microbiology from Stanford, then an MD/PhD through the joint UCSF and UC Berkeley program, an emergency medicine residency at Harbor-UCLA Medical Center, and a postdoctoral fellowship in medical informatics at Harvard and MIT. He spent twenty years as Chief Information Officer of Beth Israel Deaconess Medical Center, served on federal health IT advisory panels under both the Bush and Obama administrations, held a professorship at Harvard Medical School, wrote eight books on healthcare IT, and was elected to the National Academy of Medicine in 2020.

He had also built something once before. At thirteen, Halamka taught himself to program on an early personal computer. In 1981 he founded Ibis Research Labs — a software startup for CP/M and IBM PC platforms that he sold in 1992. That entrepreneurial instinct, dormant for two decades of institutional leadership, was precisely what Mayo needed.

What made Halamka uniquely suited for the Platform role was not his credentials alone. It was his visceral understanding of why healthcare technology had so consistently failed despite so much promise. He had watched expensive systems deployed, clinicians resist them, and transformative outcomes never materialize — not because the technology was flawed, but because the people around it had not been prepared for what it would demand of them.

“It isn't technology; technology is good enough. It's fear of change.”

— Dr. John Halamka — HealthLeaders Media, 2020

“Change management is always the hardest task.”

— Dr. John Halamka — HealthLeaders Media, 2020

His prescription for Mayo Clinic Platform was therefore not primarily technical. It was organizational and relational: build a platform that convened the right people around a shared resource, with governance

architecture strong enough to sustain trust across institutions, companies, patient communities, and regulators — simultaneously. The technology, he was convinced, was the easy part.

Strategic Decision 1: The Google Partnership and the Architecture of Trust

The structural foundation of Mayo Clinic Platform had been established six weeks before Halamka arrived. In September 2019, Mayo announced a 10-year strategic partnership with Google Cloud — a sweeping agreement giving Mayo access to cloud computing, data analytics, machine learning, and AI capabilities, and making Google the cornerstone of its digital transformation.

The partnership accomplished two things simultaneously. It built the Mayo Clinic Cloud: a Google Cloud-hosted infrastructure centralizing 1.2 million patient records in a secure environment Mayo controlled entirely. Google opened a Rochester, Minnesota office, placing engineers inside Mayo's ecosystem rather than in Mountain View — a deliberate countermeasure to the failure mode of technology partnerships where vendors understand algorithms but not the clinical workflow those algorithms must serve.

More importantly, it created the data enclave that would define the Platform's identity. The model was called 'data under glass' — a phrase capturing both its technical architecture and its ethical logic. Third parties could bring algorithms into the enclave. They could train models on Mayo's patient data. When they left, they could take only 'wisdom' — the finished, validated algorithm — not the underlying patient records. The data never moved.

“Mayo — an \$18 billion organization with 86,000 employees, and an urgency to do digital transformation — has a perfect storm opportunity.”

— Dr. John Halamka — HealthLeaders Media, 2020

The enclave's 'bin size of 10' de-identification standard ensured that any output had to be attributable to at least ten individuals, preventing re-identification of any single patient from aggregate results. De-identification of structured data — demographics, diagnosis codes, medications — ran from January through April 2020. Unstructured data — physician notes, radiology reports, pathology findings, the text capturing the reasoning behind decisions — from April through August. Unstructured data was harder, slower, and more consequential. It was also where the most valuable clinical insight lived.

Governance over the enclave was multilayered. A multi-stakeholder 'One Table' task force reviewed all data access requests. A Health Data and Technology Advisory (DaTA) Board, established in 2021 with eleven community members, provided explicit acknowledgment that data governance was not purely a technical or legal question — it was a social one. Patients whose records were in the enclave had not consented to any specific use of their data; the community board was designed to represent their interests without making individual consent logistically impossible at scale.

Strategic Decision 2: The Convener Model

Infrastructure was necessary but not sufficient. Halamka understood a second-order problem clearly: an enclave with no users was a very expensive filing cabinet. A platform required participants — and participants required a compelling reason to show up.

“You need a platform by which you can convene people — and that may mean third-party companies, academics, or government leaders.”

— Dr. John Halamka — HealthLeaders Media, 2020

“The Mayo brand is a very powerful way to bring people together.”

— Dr. John Halamka — HealthLeaders Media, 2020

The convener strategy took concrete form in three initiatives. Platform_Accelerate, launched in 2022, embedded early-stage health AI startups inside Mayo's data ecosystem. In exchange for access to de-identified clinical data and mentorship from Mayo physicians and data scientists, companies validated their algorithms against real patient populations they could not have accessed independently. The first cohort — cliexa, Quadrant Health, ScienceIO, and Seer Medical — was followed by a second cohort of seven in summer 2022. By 2025, more than 45 startups had completed the program.

The Distributed Data Network applied federated logic at a network scale. Rather than aggregating all relevant data at Mayo — logistically impossible and politically fraught — Halamka built a coalition of partner health systems that each contributed de-identified data to a shared knowledge pool while retaining full physical and legal control of their own records. Only validated models and analytical insights crossed institutional boundaries.

In May 2023, the network expanded internationally, adding Hospital Israelita Albert Einstein in Brazil, Sheba Medical Center in Israel, and University Health Network in Canada — bringing the total to approximately 27 million patient lives across three continents. The international expansion was strategically significant: algorithms validated only on American patients are fundamentally limited products. Geographic diversity in training data is a clinical requirement, not a diplomatic gesture.

By June 2024, the network had grown to nine institutional partners on four continents representing 56 million patient lives. The developer ecosystem had nearly doubled to 81 companies. Sixty-one healthcare providers were actively integrating platform-based solutions into clinical workflows. Platform_Insights, the third initiative, allowed established health systems to benchmark their clinical outcomes against the broader network — expanding the platform's value proposition beyond algorithm development to operational improvement.

Strategic Decision 3: The Microsoft Partnership — AI Inside the Workflow

The Google partnership addressed data infrastructure and algorithm development. A second, equally urgent problem remained: physician burnout driven not by clinical complexity but by administrative burden.

By 2023, American physicians were spending more than half their working hours on documentation, prior authorization, inbox management, and compliance reporting. Some estimates placed administrative time at two hours for every hour of direct patient care. The mandate to adopt electronic health records, designed to improve information sharing, had made the act of documenting a clinical encounter slower and more onerous than the paper it replaced.

In September 2023, Mayo became one of the first healthcare organizations to deploy Microsoft 365 Copilot, enrolling hundreds of clinical staff in Microsoft's Early Access Program. The focus was not clinical decision-making but documentation: using large language models to automate form-filling, summarize patient records, draft correspondence, and reduce the cognitive friction between what a physician needed to record and how long that recording took.

“Privacy, ethics and safety are at the forefront of Mayo Clinic's work with generative AI and large language models.”

— Cris Ross, CIO, Mayo Clinic — Microsoft News, September 28, 2023

The deliberate framing was strategic. By 2023, the industry had watched IBM Watson Health — once promoted as the future of oncology AI — fail to deliver on its clinical promises and be divested to a private equity firm. Mayo was not interested in repeating that pattern. The Microsoft partnership was subsequently extended into a research collaboration with Microsoft Research and Cerebras Systems, announced in January 2025, focused on training foundation models on multimodal clinical data for radiology, pathology, and genomics — moving from deploying general-purpose AI to building clinical-grade models from the ground up.

The IBM Watson Shadow

No serious discussion of Mayo Clinic Platform's strategy was complete without understanding the failure it was designed not to replicate.

IBM Watson Health launched with extraordinary ambition. In 2011, IBM announced that Watson — the system that had defeated Jeopardy champions — would recommend cancer treatment plans. Memorial Sloan Kettering became its primary training partner. Major hospital systems signed contracts. The promise was transformative: AI synthesizing the entire medical literature to recommend optimal chemotherapy regimens, customized to each patient.

By 2022, IBM had sold Watson Health to Francisco Partners at a significant loss. The failure had identifiable structural causes. Watson was trained primarily on cases from one institution and did not generalize across patient populations. MD Anderson Cancer Center discontinued a \$62 million Watson project in 2017 after finding that its recommendations contradicted physician judgment and, in some instances, suggested treatments clinicians considered clinically unsafe. Watson's training data consisted of physician-annotated cases — not actual patient outcomes — meaning the system had learned to mirror the preferences of oncologists at a single center, not to improve survival rates. Commercial pressure to announce capabilities consistently outpaced the clinical evidence that those capabilities existed.

Every structural choice Halamka made in building Mayo Clinic Platform — the federated governance, the DaTA Board, the bin-size-of-10 de-identification standard, the Platform_Accelerate validation requirement, the insistence on FDA clearance as a threshold for clinical deployment — reflected a deliberate attempt to build in the accountability mechanisms that Watson lacked. Whether those safeguards would hold at 56 million patient lives and 81 solution developers was a question 2025 had not yet answered.

The AI Question: Augmentation or Replacement?

Throughout this period, Halamka returned repeatedly to a question the healthcare industry was actively avoiding: what, precisely, was AI for in medicine?

“Machine learning is pattern recognition. As a clinician, I was trained to recognize patterns.”

— Dr. John Halamka — HealthLeaders Media, 2020

“Machine learning is not replacing humans, but offers tools that augment human decision-making.”

— Dr. John Halamka — HealthLeaders Media, 2020

In public appearances, Halamka also deployed a more deliberately provocative formulation that generated substantial debate:

“If your doctor can be replaced by AI, then your doctor should be replaced by AI.”

— Dr. John Halamka — CXOTalk, 2023

Quoted in isolation, the statement read as a threat to the profession. In context, Halamka was making a narrower argument: if a physician's contribution to a specific clinical decision is primarily pattern recognition from structured data — reading a chest X-ray for a standard fracture, flagging a threshold lab value, identifying a known drug interaction — that cognitive task can and should be augmented or replaced by AI. This frees the physician to apply judgment precisely where AI cannot: the ambiguous presentation, the patient who needs a diagnosis explained, the treatment decision that involves values as much as evidence.

Mayo's deployed algorithms illustrated the principle. The Platform had enabled tools for identifying atrial fibrillation risk from EHR patterns, flagging prediabetes before symptoms appeared, predicting 10-year colorectal cancer risk in asymptomatic patients, and reducing colonoscopy polyp miss rates to a 3 percent error level. In March 2024, the FDA cleared ELEFT — the Eko Low Ejection Fraction Tool — co-developed with Mayo Clinic, enabling a stethoscope to detect low cardiac ejection fraction without an echocardiogram. The algorithm was built and validated inside Mayo's enclave, then commercialized by Eko Health. The wisdom left the enclave. The patient data did not.

The Strategic Tensions

By 2024, Mayo Clinic Platform had reached a scale that would have seemed implausibly ambitious as a five-year projection in 2020. Scale, however, has a way of amplifying tensions that governance structures designed for smaller numbers cannot easily absorb.

Speed versus Safety. The technology and pharmaceutical industries operate at fundamentally different tempos from healthcare's clinical governance. A health AI startup can iterate a model weekly. Mayo's institutional review, ethics oversight, and validation requirements operate on timelines measured in months. Startups in Platform_Accelerate sometimes pressed against those timelines. Mayo's clinical leadership sometimes worried that commercial momentum was outpacing the evidence base. IBM Watson's trajectory was the reference point no one needed to name.

Mission versus Revenue. Mayo Clinic is a nonprofit whose mission is patient care, not investor return. Platform requires sustained investment in infrastructure, engineering, governance, and international expansion — all requiring justification within a nonprofit framework. As Platform grew commercially, it drew increasing scrutiny: from regulators questioning whether the data use required additional oversight, from patient advocates questioning whether commercialization exceeded what patients had understood when their records entered the enclave, and from within Mayo itself — clinicians who had chosen a nonprofit institution for reasons that 'digital data business' did not fully honor.

Openness versus Competitive Advantage. The distributed network model required institutional partners to trust that Mayo would not exploit the relationship at their expense. But Mayo competed with some of those partners for patients, research funding, and physician talent. The 'One Table' governance was designed to

manage this conflict; whether it could sustain trust across nine institutions on four continents, with 81 solution developers and partnerships with two of the world's largest technology companies, was untested at that scale.

Institutional Identity versus Platform Logic. Halamka had framed his mandate as building a 'digital data business.' Platform businesses operate by a logic — expand the network, monetize interactions, extract value from ecosystem participants — that sits in inherent tension with the values underwriting Mayo's brand. How far that logic could be pushed before Mayo's nonprofit mission, its 157-year institutional culture, and its patient community began to notice the distance was a question the case left open.

The Current State: 2024 and Beyond

By 2024, Mayo Clinic's financial performance had strengthened substantially. Operating income of \$1.1 billion represented an 82 percent increase over 2022 — driven by revenue growth, operational improvements, and the addition of 14,000 new staff positions. Capital expenditures reached \$1.18 billion, much of it directed to the 'Bold. Forward. Unbound.' expansion of the Rochester campus, the largest construction investment in Mayo's history.

“By every measure, Mayo Clinic made excellent progress in 2023 towards our goal of creating a better healthcare system.”

— Dr. Gianrico Farrugia, CEO — Fierce Healthcare, February 2024

The Platform had grown faster than Halamka had projected. Nine distributed data network partners. 56 million patient lives. 81 solution developers. 61 healthcare providers integrating platform solutions into active clinical workflows. At PlatforMed 2024 — the Platform's annual ecosystem conference — Dr. Farrugia placed the strategic bet explicitly:

“Platforms represent our greatest opportunity to bring real, lasting, and transformative change to our global healthcare system.”

— Dr. Gianrico Farrugia — PlatforMed 2024

The January 2025 collaboration with Microsoft Research and Cerebras Systems moved the ambition further still — from deploying existing AI tools to training entirely new foundation models grounded in Mayo's clinical data from inception. The goal: models capable of reshaping radiology, pathology, and genomics in ways that off-the-shelf large language models, trained on general internet text, could not approach.

But the questions Halamka had faced in 2020 remained unanswered. They had grown larger, more urgent, and more visible. The answers — on governance architecture, on mission alignment, on the long-term compatibility of platform logic with nonprofit identity — would need to be found before the Platform's own momentum made them difficult to revisit.

Discussion Questions

1. Dr. Halamka described Mayo Clinic Platform's goal as transforming Mayo into a 'digital data business.' Is this characterization compatible with Mayo's nonprofit mission? Where does the boundary between mission-driven innovation and commercialization fall, and who should draw it?

2. The 'data under glass' model was designed to ensure patient data never leaves Mayo's custody. What are the governance risks of this architecture at a scale of 56 million patient lives across nine institutions on four continents? Who should govern it — and who should not?
3. Halamka argued that 'change management is always the hardest task.' What specific organizational resistance would you anticipate in deploying AI decision-support tools at a 157-year-old institution with a deeply embedded physician culture? How would you address it?
4. IBM Watson Health was once marketed as the future of AI in oncology. It failed and was divested in 2022. What structural differences distinguish Mayo Clinic Platform's AI governance from Watson's approach? Are those differences sufficient to prevent a similar outcome?
5. Halamka stated: 'If your doctor can be replaced by AI, then your doctor should be replaced by AI.' Evaluate this claim. What is the strongest argument in its favor? The strongest argument against? What does your answer imply for how healthcare systems should be structured?
6. Mayo Clinic competes with some of its distributed data network partners for patients, research funding, and physician talent. What specific governance mechanisms would you recommend to manage this conflict of interest, and what are the limits of those mechanisms?
7. Mayo's 2023 capital expenditure was \$1.18 billion, directed largely toward physical campus expansion. How should Mayo weigh physical infrastructure investment against digital platform investment? Does a tension exist between these strategic directions, or are they mutually reinforcing?
8. Platform_Accelerate grants startups access to Mayo's patient data and clinical brand in exchange for validation work. Who captures more value from this exchange — Mayo or the startups? Is the exchange equitable? Should startups receive equity stakes, revenue-sharing agreements, or other consideration for the commercial value their validated products add to the Mayo ecosystem?

Notes

1. Fierce Healthcare. 'Mayo Clinic closed out 2023 with \$18B in revenue, 6% operating margin.' February 2024.
2. Wikipedia. 'John Halamka.' https://en.wikipedia.org/wiki/John_Halamka [verified April 2026]
3. HealthLeaders Media. 'John Halamka, MD, Launches Mayo Clinic's Digital Data Business.' 2020.
4. Mayo Clinic News Network. 'Mayo Clinic selects Google as strategic partner.' September 2019.
5. National Academies of Sciences, Engineering, and Medicine. 'Sharing Health Data: Chapter 12 — Mayo-Google Partnership Case Study.' 2023. <https://www.ncbi.nlm.nih.gov/books/NBK594445/>
6. Mayo Clinic News Network. 'Mayo Clinic Platform expands its distributed data network.' May 2023.
7. Mayo Clinic News Network. '4 Takeaways from PlatforMed 2024.' June 2024.
8. Microsoft News. 'Mayo Clinic to deploy and test Microsoft generative AI tools.' September 28, 2023.
9. CXOTalk. 'Data and AI Improve Patient Outcomes at the Mayo Clinic.' 2023. [existence confirmed; full transcript not publicly accessible]